

ActInf Textbook Group ~ Cohort 1 ~ Meeting 11 (Chapter 5 pt. 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 11 (Chapter 5 pt. 2) | 57:54

all right hello everyone it is week

11.

and we're in chapter five part two

of the active textbook group first

cohort

will be now turning to some of the more

direct questions on chapter five after

having more of a overview

last week

we'll also

um

be concluding our regular chapter

discussions

for the first part of this book

and in the coming two weeks we'll be

having some review and synthesis

and connecting some dots

asking overarching questions returning

to questions that were based in a

specific chapter and hopefully people

like on their own or wherever else

they're seeing is relevant

or requesting new pages if they see

something that they want

providing some synthesis on this first

half of the textbook which is the

epistemic half and then

um for people who want to continue

in the textbook group

go to the onboarding page

and there's two columns there's yes for

part two of cohort one chapter six

through ten so check that box if you

want to be included in starting in

September

going through chapters 6 through 10
which has like the recipe for active
models and that's when we're going to be
getting more Hands-On with the modeling
also everyone is welcome to join for
part one chapters one through five
cohort two also starting in September
so it could be

um a fun experience just to get another
coat of paint on the material
play more active role ask another set of
questions informed by what you've seen
in the first cohort so feel free to um
check those boxes

and then also

[Music]

um

keep incubating as we head into chapter
um six and Beyond in the second half of
the textbook

um like the project ideas several
um of the project ideas like related to
what Mike and Noah have added we've
started to address and explore in the
active block runs meetings on Wednesday
especially related to like cyber
physical systems simulations
taxonomies for governance
these kinds of ideas

um Brock's added some great notes about
like what what is the ramp that gets
someone to the low road or the high road
um and then uh I've recorded the first
chapter

and the first segments as an audiobook
the first chapter doesn't have any
equations so it's
straightforward to read but the

subsequent chapters do have equations
which is um
motivating
having the natural language
representations of the equations
so there's still a ton of
um discourse questions that can be asked
why things are one way why things are a
different way if it can be placed into
the equation itself that's great if it
needs to be somewhere else hopefully the
space exists and people can request if
not but
um for example when reading the textbook
the equations can be read this way so
those are just some of the projects
that people can look forward to and the
affordances to continue on which is
again uh going to the onboarding and
participating in first five chapters
again cohort two or going to continue on
with chapters six through ten starting
in September for both of these so that
we can like complete one pass of the
book by the end of the year and also
probably make a lot of progress on
different project ideas and project
ideas arising as we start to see and
build templates for the models
and also like Ali has some nice
interactive notebooks so there will be a
lot of great things um coming together
if anyone wants to raise their hand of
course just go for it
otherwise we will be continuing with
the
questions asked about chapter five
before we go to the questions is there

any general comment that somebody wants
to add about
chapter five the reading of it they're
rereading of it listening to the video
from last week
how is chapter five
um
hitting them today
okay
um to kind of maybe bring that one step
closer to a specific question
I'm not going to
read this whole quote but this question
says in the preface
they wrote in chapter 5 we will move
from formal treatments to biological
implications of active inference
this AIDS in mapping the abstract
computational principles of active
inference to specific neural
computations that can be executed by
physiological substrates
this is important in forming hypotheses
under this framework and ensures that
these are answerable to measure data in
other words chapter 5 sets out the
process Theory associated with active
inference
is this how people
saw chapter 5
do they feel like it did something
different did it succeed
at making the abstract computational
principles linked or mapped to specific
neural computations executed by
physiological substrates
yes Eric
you know first of all I have to

apologize I only watched half of last
year's last week's discussion
so I may be cut talking about ground
that's been covered before
um it just seems to me that you know
I've seen many many

I
um papers
um trying to map theories onto brain
circuitry and physiology
and it's

I I'm not well enough
um you know I spent you know I'm only
loosely associated with neuroscience and
you know

don't follow it very closely so it's
hard for me to cite specific instances
along the way I know you mentioned
numanta which is um you know one of the
more recent ones it's
trying to say this is these are what the
layers are doing

um so
this chapter to me didn't seem
um any very qualitatively different from
anybody else's very hand-wavy way to say
here's what the computations are all
about and here here's how they map to uh
what brain circuitry could map to what
base circuitry might be doing it's all
quite hand wavy I I find and

um
I guess plausible but um
it's certainly not definitive nothing
close to that so

um
maybe one other
one other comment I would say is um

again in this chapter they use this word

prediction

in a way that conforms to the theory

but to say that you know what what's

happening in

um say

um motor control where top-down signals

from the cortex go down to the ganglia

in the in the spinal cord in order to

invoke motor patterns

um

to call that a prediction is to me a

kind of um well that's that's taking

Extreme Measures to map the concepts of

the theory onto

what neurons are doing

and um you know if the theory works out

to be

you know very useful in some way and

that's what you have to do to to make

that happen

okay I guess I can use that language but

it operatory it just does not seem like

the right language to use to say yeah

we're trying to to set up some motor

patterns set up some some context and

situations for what we want the you know

the local reflexes and and to be doing

um but to cast that as a prediction just

seems a little unnatural to me

so those are my two reactions to this

chapter

thank you

anyone else with a raised hand

this is a really

good point about what

the linkage or mapping is between

uh without going into like you know is

anything physical or Etc like
physical brains in their Niche and any
type of equations
would a well-fitting linear model say
that the brain is a linear model
does a well-fitting base graph imply
that the brain is a base graph
does a well-fitting base graph plus a
philosophical belief that we're just
making a map we're not describing the
territory equate to what

Beyond a linear model or a verbal model
Rohan

yeah building on the previous point
right uh

I am still unsure what what uh active
inference is considering a signal
there's no effective
well there's no noise model or there's
it it doesn't specify that here's
because if you if you say
uh that I'm predicting something right
so there must be something that is
substantially different statistically
different from some noise distribution
that you can differentiate right so
it's basically trying to estimate some
hidden probabilities which is not really

Sigma

does it make sense because yours even
the internal States unless they're
actually

uh instantiated prior like when we set
up the model we say okay these are the
only states that you can actually

Traverse

it does not make you still have to infer
the States from your measurements of

reality

thank you

foreign

yeah I just

um

wanted to ask if the noise isn't kind of
implied in

um the pomdp formalism

like uh I think in figure 5.4

um

where it shows the direct and invert
Pathways with message passing

um

it describes

active in the basic ganglia

as a umdp generative model so if it's
partially observable doesn't that imply
some kind of noise

uh my counter to this would be uh that
is missing information that needs to be
filled out that's not really noise rate

uh so noise would be something

like if we

uh like it so even if we let's let's not
use this right it would be some so we
have like returns of a stock

and looking at the distribution of of
the prior returns we know what the
spread is and in case we we want to
measure whether your portfolio is
substantially different from this
distribution that's where we would have
some sort of like sharp Ratio or a
signal to noise ratio noise ratio being
the complete set of distributions that
are possible

this set of ranges that are possible

uh that is not really here because it

that is missing information so we can only observe part the fact that we have only partial observations does not excuse the fact that we don't have uh even noise models of the sensors so how good are the sensors for example I do we have some idea of that yes but then I'll add a point about these Bayesian terminology yeah um

yes um is there is there any like what's the Practical difference between partial observation where there's where the noise implies some kind of information that is missing versus like a distribution that's added to full observation that

um adds noise to like the sensory processing great so it so that would be some okay so if I use an actual physical sensor right

uh it would be something like what are the ranges of values that it can go up to because we've designed the sensor so we would we basically know what the bias is what the variance of the sensor is under different conditions

uh there are actual physical processes implied in that measurement so in the case of uh

ultrasonic sensors probably like some current value or some voltage value that's being measured by some circuitry somewhere so there are ranges for that

so it there's no missing information

there

so you basically know

so it's already prior in this system

right so you know what the ranges of

values are going to be and from there

you need to infer the some state in the

world so is there an obstacle in the way

or is there yeah

okay let me just make a comment about

some basic terminology then um Jessica

and Brock so let's look at figure

4.3 this is the partially observable

uh Markov decision process that Jacob

talked about but we're going to be just

talking about the perceptive part not

the action intervention into how things

are changing get just focusing on this

Motif on the left like step-by-step

guide uses

so

um you mentioned about the the bias

related to measurements so if

the measurables are the observable sense

States then the bias and the covariance

and the variance

about how like the reading on the

thermometer relates to the underlying

variable the the hidden State the

temperature of the room it's embodied in

the a matrix

because that is actually the learned or

fixed variable

that is

playing a role of a noise model

then you mentioned when talking about

Returns on stocks testing whether

there's like a substantial difference

from some null expectation like is there
a signal that's different from the noise
um this is like a frequentist approach
it's based upon determining whether
there's some statistically significant
level doesn't have to be it's not the
only way to do it

um but one can do like a p-value on
whether a given signal is statistically
different from given a noise model
different from noise and then like the
sharp ratio and other types of things
are like summary statistics they're like
descriptive statistics

um

noisy measurements

are addressed in the context of the
pomdp

by having

an a matrix that accommodates some of
the features of denoising a signal
and then Precision variables
and those precisions can again be
learned or fixed

but like the functional

um

aspects of what is being referred to as
noisy measurements

are accommodated within this kind of
Bayesian approach which is why common
filtering generalized Bayesian filtering

why these techniques are used because we
can have a Bayesian update scheme
implemented with with variational
approaches message passing sampling all
these things

that does denoise signal and it does
embody a noise model and a signal model

and that's all part of the generative model of the cognitive entity so it just isn't being addressed from a frequent to statistics or um

yeah I shouldn't have brought up the sharp ratio maybe that's not the best but the sharp ratio is uh is effectively some way to say that some portfolio choice is different from others or in communication systems you would see use something like EB over n_0 that's the signal noise ratio for any sort of communication system so yeah yeah uh but they're not necessarily only frequentest right so we there are patient ways to estimate these things in fact that's pretty much how we do it in practice because we we don't assume uh that we know the uh complete distribution we update priors in each of these cases because we have yeah uh

coming back to your point that the a matrix does address this the a matrix would be something that would you said it would denoise and filter out some stray observations but okay so our so the the question Still Remains is when you're saying prediction is a prediction of the signal over here as in what is the future state of the sensor but I think there it seems seemed to indicate that the prediction relates to relates to the free energy minimization

itself right so you choose policies that
would minimize your free energy and keep
yourself into some set point
so which means that there should be some
sort of uh
an idea of what ranges are there for
this value or uh how far can these go
before
you know you're not in that set point
anymore
so even if you if you do want to make
that prediction it's a it's not just the
signal value so signal values
effectively through the Marco blanket
they affect some internal State and
we're that's what I I'm assuming that's
what we're trying to predict the
trajectory of the future internal States
actually the expectation that's being
um yeah minimized is over outcome
observations not about internal States
oh okay yeah yeah I'm sorry so uh yeah
so even then we we don't have so the
signals may have well-defined precision
and bias but
the entire system itself doesn't have
any sort of
uh
well it doesn't know how wrong it could
go
it doesn't have any error estimate so
that it's it there's no noise model as
such so we can't apply any so we would
have to assume that there is some
distribution of states and then use
something like least squares or some
other uh optimization routine to
actually bring this in practice

great totally possible um it would be interesting to sketch out a Bayes graph for what you believe a signal to noise or some type of noise model would embody and maybe we will see some analogies or some mappings

um Jessica and then Brock

hi yes I'm I mean I think

I both you and Rohan like in the new Province but

I was going to say that um

would be like the filtering and like a lot of like the biases and things like that kind of like made sense of it is like

I see the because I think it was like chapter two or something like that I said like right like so the why is

this you know quote-unquote

um you know uh like a fat thing like you know that we can observe from the hidden state or like the process of something that we don't know and so for everybody like we can see the same thing but how we process it internally is going to vary like it's gonna be very subjective so the why is gonna look different for Daniel then it's gonna look for me based on all this filtering and so the black Estates

I imagine sort of like one half like those filters that encode the biases and all the things that allow us to process things differently and and so like then I generated model and it would like it's gonna be updated like differently all right also like encoded I think in the general model

would be a lot of uh like you reset and things like that that embody I think probably like a lot of biases and like things of how we understand things which also influence how we interpret what we observe and so for so for me it's like a lot of these things are there like in terms of like statistics I don't know um much but that's how like I you know conceptually I um

but like you know understood like a lot of these things so like basically like in the blanket stage of the Marco blanket

it would have a ladder like this filtering you know and yes like

um like if you're going to try to you know let's say Like You observe something that's different than what you predicted in your model then you would take action and to forage and like find information

and to figure it out okay like what am I missing like why did I make this mistake or like this error I in my prediction and so that I can like have more information and update the model so that it aligns more closely so then you would basically take action to the noise and basically get closer to

um like quote unquote the reality that's out there

that yeah I don't know if it's official lighting but that's how like it kind of makes sense to me

thanks Jessica Brock

um yeah I just

uh

wanted to I guess speak to this like

uh sensor noise model I don't like in

practice in reality like when you're

talking about sensors like in a physical

system like like a digital signal

processing kind of context like

the sensors are tested under some

conditions for things like you know

Jitter and

um spurious kind of uh noise conditions

and stuff like that but they're tested

under some conditions

meaning that there's never ever going to

be a time when you have an actual model

you're just going to have an

approximation whether that's expressed

in a frequentis or Bayesian

um

way like there's going to be some

uncertainty in the model

and especially if you start connecting

it up to

a larger system

um doing the same kind of like is the

Jitter in that sensor the same when you

know when it's completely isolated as it

is when it's on a you know six by nine

PCB board with you know

um

12 volts on one side and a oscillator

next to that that's you know like

are those now is it the same thing you

know

um you have to do these sort of

electromagnetic compatibility you know

resonance sort of testing on it like

there's just that it's it's uh in
practice it's Bayesian like you're doing
the same sort of
um bounding and you know belief
um testing and you know Finding Your
errors and correcting them
um even in practice

[Music]

um

I don't see how you could possibly have
like even like physically a system like
that could exist where you had a noise
model that you could write down that was
actually the noise model

um under all conditions or

yeah it's um

it doesn't seem like a
approachable I don't know it's not even
not tractable

which is the wrong thing to do I don't
thanks Brock

um in the case of engineering and
designing models it's up to the designer
or the engineering team to understand
like what is adequate or not and for
natural systems

outside of their bounds

natural selection sweeps them off the
table

so some of the adequacy questions are
either based upon

specified or implicit human standards or
the failure to resist

dissipation

and therefore the failure to realize
repeated measurements

of oneself or from the external

and then that system is no longer

let's just continue to move through some
of the questions

um yes go for it

uh yeah I I completely agree with what
Brock said right but my my point was
when when you say that you're predicting
something it means that

uh you either know the quantity that
you're trying to predict
and or and it corresponds to so in this
case like some action results in some
reward I guess you could predict the
reward but without knowing how wrong
you're going to be in the future

so

where exactly immigrant is so that any
living system would have to have be
aware of its own bounds right that's
that's what I meant by noise models so
it has to know that it has to stay
within some bounds and as it starts
coming closer to one of the upper lower
bounds there should be some something
like pulling it back down

or some feedback mechanism but that is
not very clear from the from what I've
read so far on active inference system
what

how exactly it would estimate that is
you can just save free energy
minimization

uh but what is a minimum in this case
that would be different for different
systems so it should have some idea of
where the surfaces of whichever local
surface it's lying on that that was my
point

so for sure it's different for different

systems it's different as every single generative model is

um and then you mentioned like

um a pullback attractor and so this has been treated extensively in the context of physiological measurements and then again wear those priors on the physiological measurements are the tolerable range are either provided by the human engineering team or through evolution by natural selection and then

um yeah

the quantity like another related topic would be expectation maximization models which is what is happening essentially in the dialectic between the observation and the hidden State update is an expectation maximization like process where it's like given the hyper priors

or we can just simplify just given the priors on S what are the most likely or the distribution of observables and then given the observables what should be updated about the hidden State priors and when those are at conversions the model is stationary and then as things change either from the top via learning or context shift or through changed measurements

the expectation maximization algorithm is just able to track those changes and that can in the context of a fixed prior have the function of a pullback attractor

um okay and that's even before getting into action specifically but yes definitely considering like the role of

comparing future plans of action
and which plans of actions will have the
least expected free energy based upon
their
um pragmatic and epistemic value like
which the epistemic just being Clarity
around how it's going to be achieved and
the pragmatic value being that KL
Divergence between the expectations
preferences and the expected
observations
so if we expect slash prefer to be in
homeostasis
the pragmatic value is going to come or
be loaded onto policies that keep us in
homeostasis even if some other one is
super informative
how that model gets tuned is quite
literally the details of how the model
is
trained right my point was that active
inference doesn't help us find this we
have to find this ourselves and then
hopefully once we have instantiated
these bounds I think active inference
works very well to keep it within this
right if you can't discover this you
know right like I can't have like a uh
what you'd say double or rasa type
system like a blank slave system God in
the world and discover your bounds
that's not going to happen but this kind
of method is what what I was uh making
it out whereas
taking some something equivalent without
any so deep learning is another
uh method right so you just feed it a
lot of data it doesn't have to know

anything about the data eventually it
forms some opinion about the data it's
seen and then it's able to do some
classification or depending on how much
data it's seen so it's able to build
certain amount of bounds whether that's
generalizable is a different question
but

uh here there's the learning part
does not say much about how exactly it
goes about discovering
where its limits are that that's my
point

sure thank you

um Eric

um you know I would I would love to to
dive this a little bit more into

um the thing you were just talking about
how expectation maximization might is a
is a mathematical

formulation for how to infer belief

States from observations because you uh
basically go through this iterative
process of trying to find alignment
between the states of the belief and the
observables and then you can track that
over time

between that concept and the concept of
message passing

when I think of message passing I think
of some sort of a distributed system

where you have some local computation
and it sends you know and you have again
either synchronous or Asynchronous
process for

communicating between these kind of
different centers of
cut of belief state

and then again through message passing
you have often an iterative process of
convergence

so um those are not the same thing and I
think you know message passing to my
understanding would be an em algorithm
under certain types of messages and
States at each of the nodes that you're
sending the messages around but not all
message passing is going to be any em
algorithm and similarly when I think of
em I don't think about message passing I
think about making an arrays of you know
of um

belief and matrices between you know
mapping between belief and observations
and then

um trying to compute an expectation for
you know for for what my predictions
would be and then making this iterative
conversion so I don't think of that in
terms of message passing so I'd like to
I just love to hear other people's ideas
about how to tie these two
um ideas together

yeah

um Brock

if your hand is still raced or

okay nope so sorry yeah okay all good
um

does anyone have thoughts on this
there's a lot to say about the uh
expectation maximization algorithms and
like just to kind of give a little
context this is the step-by-step
um paper
so this is the model stream one it's
four parts

um and this is like really a quite relevant figure which is not in the textbook but helps a lot like this is the essence of the partially observable Bayesian inference there's a prior D and then there's an ambiguity Matrix that is mapping in a generative capacity between hidden States s and observation measurements o this is The Tail of Two densities because o through a can give you the most likely s and S through a can give you the most likely o and it turns out that by alternating those procedures in a Bayesian update context that two-stroke engine is called expectation maximization again because given expectations on the summary statistics of a distribution the expected sensory observations can be generated and then given those observations a likelihood function can be maximized that then updates the hidden States so this Motif this kind of elbow motif is then extended into a caterpillar with this B which is how the hidden State changes their time so importantly like note what's not there which is like the observations being chained Through Time so it's not that the temperature reading at one moment influences the temperature reading at the next moment the temperature readings are continually

linked

um synchronously like in one time slice

to their hidden States

so then when action gets into the

picture

we've talked about like several sources

that's like the equation 2.5 to equation

2.6 phase change because action brings

in several kinds of uncertainty first

off you're reducing your uncertainty

about observations which haven't

happened yet

you're also having the unknown

consequences of your actions

and in order to have action selection

that's relevance one has to have a

preference distribution

over what kinds of observations they

would like to be seeing so that action

can guide it in that direction

okay

um then so that's the the expectation

maximization two-stroke engine

is basically graphically

um proposed here D is just the

initiating set of parameters

and then in a One-Shot way like you can

have a a folder with a thousand images

and do expectation maximization on it

or you could have something that's

dynamical and do that type of

expectation maximization Through Time

[Music]

um

Eric asked how is it related to

message passing

on graphs so I'm not familiar with all

the details of this model but you

basically said it which was that like
certain message passing systems

Implement em

but of course not all message patching
architectures Implement DM so message
passing is more General than em also em
wouldn't have to be implemented through
message passing so there's kind of but
there's like an area of intersection
where the EM algorithm can be seen as a
type of message passing

under certain compute rules in a factor
graph

and EM at least from here may be used to
break Cycles in a factor graph

so it's always like

how close to the kernel and like the
platonic ideal of active inference are
we talking

and then how many heuristics
and just

ways of connecting things are possible
so I

I will now read the chat

that's from previous context okay okay

thank you

um

like the statistical parametric mapping
textbook and documentation SPM
has a lot on em
variational inference

and a lot of the parts that bring one
to understand this

a lot better SPM is like almost like it
doesn't include action that much because
it's a neuroimaging so the observations
are neuroimaging Sensor Fusion different
error modalities like the whole panopoly

and then SPM started to incorporate
participant actions
in this partially observable metabasian
way and potentially that is what led
Tristan colleagues towards a grander
synthesis of inference and action
under a statistically principled
framework

let's just see if there's any other
these are all you know
important questions and like there's a
lot to get up to them and then there's
like a lot to go from the question but
these are all important things to raise
okay

[Music]

um
if anyone has a thought on this how
should we think about redundancy in
neural systems in the context of active
inference

so redundancy would be like if there's
something that's playing a functional
role but it's removal
is not damaging the function of the
system yeah there's 10 pillars knocking
out one of them the building stays up
so there's redundancy in the pillars
what about the cases where the same
neural circuits serves multiple
functions

so how do we deal with the fact that
there's like a many-to-many potentially
or a complex mapping between system
elements and system functions
where sometimes removing Node 1 does
nothing but removing node one and two
there's a lot of things but then

removing one two and three and it's
going back to being fine Jessica
and maybe like ask blue about it because
this is what I feel but
it this question made me think about
like it's not only like that the brain
has like this redundancy things but
it like would like to say like if you
have some kind of damage to your brain
like another part of your brain that
maybe you was not
use for something would develop that
capacity
and so it would create that you know
ability
and so neuroplasticity like
brand new even though it was not a
redundancy to begin with so I guess I'm
more Curious to understand this and so
other people who know more like blue and
stuff um
about this and basically that's kind of
what this question made me think about
and kind of wanting to understand more
that's good question yeah blue wrote
depends on your age yeah like difference
neuroplasticity mechanisms are
differentially available throughout life
for different organisms
um
Just one thought on this would be like
the same neural circuit serving multiple
functions might be totally the case for
example
um and one like is the function of a
heart to provide one pound of weight to
the Torso it is a function of the heart
is at one you know so identifying what

function is being modeled is what's
being done here so let's just go to like
the example of figure five three
this
neuron
modifying it like a loss of function
experiments or having it injured might
influence more things than just
the lower motor neuron descending
message
so that'd be a case where like it serves
multiple functions
um
this is just one statistical model
of
this function of this circuit
and that returns to to the earliest
comments of Eric with like okay this is
like tantalizingly seeming like it's
actually going to be describing the
neural Anatomy but then they say things
like note the absence of dot dot dot dot
note that there's a discrepancy between
for example
the reality of the anatomy
and the base graph
this highlights that the connections
implied by message passing schemes may
not manifest as single synapses
so
okay
there's kind of false positives and
false negatives the Bayes graph isn't
just the anatomy
that's fine
the linear regression between height and
weight isn't the actual relationship
between height and weight and the

structural equation model of
inequality is not the generator of
inequality
so it's totally fair and shouldn't be
expected to be otherwise that the base
graph the best fitting base graph the
most didactic phase graph the simplest
phase graph
none of those recapitulate the anatomy
and the base graph that recapitulates
the anatomy would not necessarily even
be the best fitting
its map and territory
and if somebody has a special equation
to break through that
blanket
everybody would love to see it
however
sometimes it's easy to gloss over
that
in principle challenge
of mapping formalisms to biological
systems because we see the cell and then
the blanket is the membrane and we see
the brain and it's like vision and
action and it seems like it maps on to
the physical or the anatomical structure
of the world or the causal structure of
the world
and um
it doesn't
so I hope people can add more thoughts
about like how we consider redundancy
because it's a great
question
um
but we'll leave it there
um how are

reflexes modeled in active inference
would they exist in an active inference
module that is distinct from
proprioception but they operate on
different time scales

um if anybody wants to add like some
some context on it

um

a reflex arc

relating to

um where reflex is defined as

um a function that is being relayed

through the spinal cord of a mammal

and not passing like you could have a

nerve block in the cervical vertebra and

it still is able to like Implement like

that's one definition of reflex

often it's also used slightly more

broadly to mean like

um stimulus action reproducible outcomes

but not every reflex has a perfect

reproducibility et cetera et cetera so

it's kind of a Continuum

um yes it would involve proprioceptive

input

that proprioceptive input

would be combined or

um

juxtaposed with the descending

prediction

resulting in an error

and then that directionality and

magnitude of the error in this model

drives the reflex

there can also be multiple time scales

but we haven't seen any nested modeling

yet

this is just the one layer model

the schematic on the left in figure five
one
shows that layer 5 of Cortex projects to
spinal pyramidal neurons and this can be
interpreted as a prediction

okay so this is this one
um

okay helpful um context
and and it um relates to this earlier
highlighted section
which is like

it's the interpretation
the interpretive link
between some anatomical or biological
feature phenotype
and some parametric
resonance with a model

the validity of that ranges from
pretty clear
pretty uncontroversial pretty useful
pretty effective

to
none of the above
and it's hard to know like what given
interpretive links
are doing
like saying that this Edge on this base
graph can be interpreted as a descending
excitatory connection

or that might be used to generate
specific testable hypotheses like if we
measure it during a period of excitation
the activity of this neuron we expect it
to be increased whereas if it was a
descending inhibitory connection

Etc

um

that's a great example of active

inference being used in a proactive way
to generate hypotheses about biological
systems that are going to be as they
said in the preface

[Music]

um

answerable to measured data
by grounding in
the computational model
that is related to the neural Anatomy
but isn't trying to be like a digital
twin of the neuroanatomy
we can generate predictions about gain
loss of function different measurements
to make if they don't already exist
and then that can be used to increase
our confidence or reliability or falsify
even a specific generative model as
proposed

so there's so much discourse like
continuing to the present day and surely
Beyond as like all these questions and
recently posted in the Discord about
like

active inference or free energy

principle isn't falsifiable

now the funny link is it's not

falsifiable so it's been falsified

it can't be falsified

so it's incorrect

that's an interesting connection that
some people make but a linear model
cannot be falsified a base graph cannot
be falsified once any given linear model
is presented

in a context with constraints for a
certain data type

then it's an empirical question of its

accuracy and adequacy relative to other models

but at the abstract level of a neural network or linear model or active inference

falsification

simply doesn't apply

but there's a lot that could be clarified

um

about like what are the utilities and some of the pitfalls of mapping

biological or cyber physical systems to active inference models

if it's a purely digital system or potentially even a cyber physical system that's been designed a really certain way

it might be compatible with actin
like essentially by Fiat or by Design
one could make an artificial creature like an infra ant

that is implementing active inference
one could also model an ant in the field using a linear model or a levy flight or some other model or an active inference model

but those are two very different settings one in which sort of the rabbit was placed into the Hat

and then we have a rabbit analyzer and the other one being like

we don't know what's in the Hat but we have a rapid analyzer that we think applies to what is in this hat

um

chapter five includes some

um evidence presented on different neurotransmitters so
um it'd be interesting to see like what other Frameworks are able to be compatible or even provide unique explanations predictions Etc related to neurochemistry have neural networks ever been used to derive unique predictions or found compatibility with neural hormones neurochemicals
um
and then again they have a final figure in chapter five that's like a graphical overview of several of the systems that are described in the chapter specifically relating to like the cortical cognitive functions and then the dialectic between habit and free energy driven planning here with dopamine as a Precision modulator with like extensive further modeling presented in other papers and this like even more basal motor selection mechanism based upon reflex arcs and proprioceptive error minimization and all of that
um
we raise these questions not to offer any answers but to highlight some of the exciting Avenues of future research and theoretical neurobiology so a lot of the questions that are asked

um

here in this paragraph but also above at
the very beginning they said we're not
saying how it is we're just giving like
the current process model understanding
and then especially at the end
they raise even more speculative
questions

so

um that takes us to the end of chapter
five

that's the first half of the book that's
the uh epistemic component of the book
well a lot of it is epistemic but that's
the first five chapters

um

the second five chapters

which are longer

partially because of having more figures
and things like that but you can see
there's more pages

um slightly in the second

part of the book though we've also read
the um

at least some of the appendices

that's where we're heading which is
picking up on uh

chapter six in just a few weeks with a
recipe for Designing active Emperors
models so for those who stuck with the
uncertainty

for the first five chapters and stay in
the game

more will become clear when the recipe
for the dish is seen and then more will
be

clarified when the dish is prepared and
then when

you cut the vegetables and then when you design the dish in the recipe and all of that and that's like a journey that we're all going to be on the tools are not finalized in the kitchen is not completed Etc et cetera et cetera so it's just a call for us to go to onboarding and indicate our interest in continuing with one or the other uh cohort or sharing with any colleagues who you think might like to jump in to the second cohort of part one and then in the coming two weeks we'll be able to take a step back look over the chapter questions but also think about more general questions that we're having basic questions miso questions Advanced questions research Avenues and also turn to the project ideas where multiple of these are already active and there's spaces for people who want to facilitate or catalyze some other direction so fun meeting everyone's welcome to uh stay on for tools or head up to a room above if they just want to talk to other people about anything else so thanks again for joining and see you in just a few minutes for tools